

Simulation Portfolio: Multiphysics Modeling for Wireless Neural Interfaces

Drystan JUSTIN

Mechatronics & Microsystems Engineering

INSA Hauts-de-France

Valenciennes, France

Email: justin.drystan@gmail.com

LinkedIn: linkedin.com/in/drystan-justin

Abstract—As a Mechatronics Engineering student specializing in Digital Design and Microsystems, my core ambition is to bridge the gap between hardware engineering and neural interfaces. With a strong foundation in control systems and upcoming research experience in medical robotics, my long-term goal is to pursue a PhD in Bio-Mechatronics and Neuro-engineering by 2028. This portfolio highlights my self-driven initiatives in Multiphysics Simulation (Ansys), specifically focusing on the critical challenges of neural implants: Wireless Power Transfer (WPT), magnetic coupling optimization, and biological thermal safety.

Index Terms—Multiphysics Simulation, Ansys Maxwell, Wireless Power Transfer, Neuro-engineering.

I. PROJECT 1: ELECTROMAGNETIC MODELING OF WIRELESS POWER TRANSFER (WPT) FOR MICRO-IMPLANTS

A. Context & Objective

One of the major bottlenecks in wireless neuromodulation, such as optogenetic micro-LED matrices, is delivering power safely through biological tissues without heavy batteries. The objective of this preliminary study is to model the magnetic coupling between an external transmitter coil and a millimeter-scale receiver implant using Ansys Maxwell 3D.

B. Theory & Simulation Setup

Wireless power transfer for medical implants relies on inductive coupling governed by Faraday's law of induction. The absolute performance indicator of the system is the coupling coefficient k , defined as:

$$k = \frac{M}{\sqrt{L_{tx}L_{rx}}} \quad (1)$$

where M is the mutual inductance, and L_{tx} and L_{rx} are the self-inductances of the transmitter and receiver coils, respectively.

The simulation environment was configured to solve high-frequency electromagnetic fields to maximize the induced voltage in the micro-coil:

- **Software:** Ansys Electronics Desktop (Maxwell 3D)
- **Solver:** Eddy Current (AC Magnetic) at 1 MHz
- **Key Parameters:** Inductance matrix extraction across a 15 mm air gap simulating human tissue depth.

C. Geometry and Meshing

The geometric model consists of an external transmitter torus (radius: 20 mm) and a receiver torus (radius: 3 mm) separated by a 15 mm gap. Both are assigned copper properties.

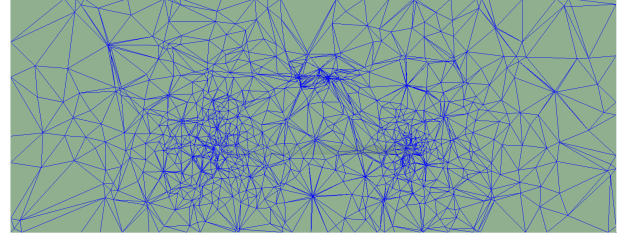


Fig. 1. Cross-sectional view of the finite element mesh generated in Ansys Maxwell 3D, showing adaptive mesh refinement around the coils.

As shown in Fig. 1, the mesh density is heavily refined near the receiver coil and transmitter edges to accurately capture sharp magnetic field gradients while optimizing computational efficiency.

D. Results & Next Steps

The simulation successfully established the magnetic flux linkage between the macroscopic transmitter and the micro-implant. To visualize this extreme spatial gradient, a dual-scale logarithmic mapping of the magnetic fields ($\mathbf{B}$ and $\mathbf{H}$) was employed (Fig. 2).

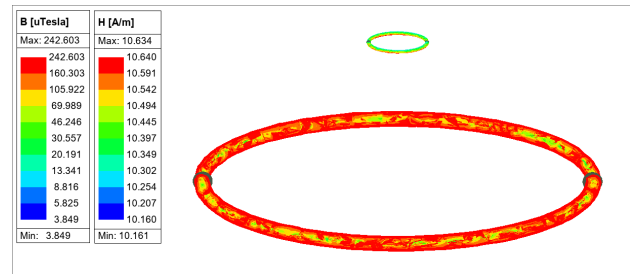


Fig. 2. 3D distribution of the magnetic fields. The magnetic flux density (B) is plotted on the transmitter coil (bottom), while the magnetic field intensity (H) is independently plotted on the micro-implant (top) to clearly visualize the coupling across the 15 mm gap.

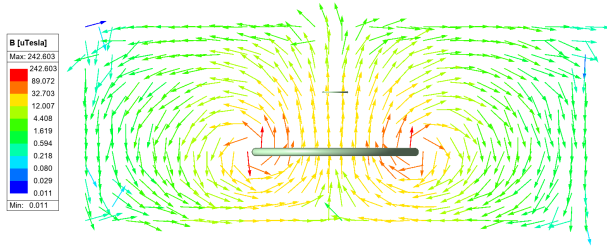


Fig. 3. Vector distribution of the magnetic field density. The logarithmic scaling reveals the flux lines successfully penetrating the micro-implant at 1 MHz.

The finite element analysis yielded a coupling coefficient of $k = 0.01165$ at 1 MHz. This value is highly characteristic of loosely coupled biomedical inductive links, confirming the feasibility of power transfer while highlighting the need for rigorous thermal management.

The next iteration of this model will integrate Ansys Icepak to calculate the thermal dissipation (Joule heating) within a biological tissue phantom.

II. PROJECT 2: THERMAL DISSIPATION AND BIOSAFETY ANALYSIS

A. Context & Objective

While wireless power transfer provides a solution to battery limitations, it introduces a critical biological constraint: thermal dissipation. The human brain is highly sensitive to temperature fluctuations. According to FDA regulations and neuro-engineering standards, the maximum allowable temperature increase (ΔT) in brain tissue is strictly limited to 1°C to 2°C to prevent irreversible thermal necrosis. The objective of this project is to simulate the continuous thermal footprint of the micro-implant operating at a 50 mW power level.

B. Theoretical Framework

Heat transfer within biological tissues is classically modeled using Pennes' bioheat equation. For a steady-state, worst-case scenario analysis (neglecting the convective cooling effect of blood perfusion), the thermal diffusion is governed by the 3D Poisson equation for heat conduction:

$$\nabla \cdot (k_t \nabla T) + q_v = 0$$

where T is the tissue temperature, k_t represents the thermal conductivity of the biological medium (approximated as water/brain tissue, $\approx 0.6 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$), and q_v is the volumetric heat generation ($\text{W} \cdot \text{m}^{-3}$) stemming from the implant's Joule heating.

C. Simulation Setup

The thermal simulation was conducted using Ansys Icepak. The boundary conditions were established to mimic the human body's thermoregulation:

- **Tissue Boundary:** A Dirichlet boundary condition of 37°C was applied to the outer faces of the simulated brain tissue domain.

- **Thermal Load:** A solid heat generation of 50 mW was assigned to the receiver micro-coil, and 0.1 W to the external transmitter.

D. Results & Clinical Implications

The steady-state thermal mapping (Fig. 4) illustrates the spatial temperature gradient diffusing from the micro-implant into the surrounding tissue.

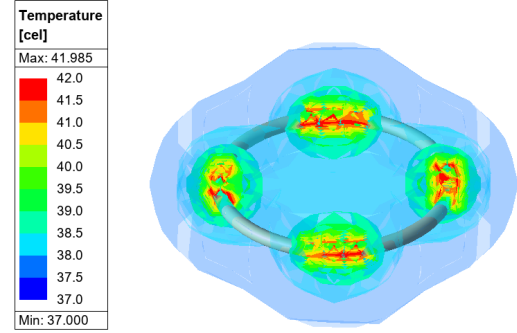


Fig. 4. Steady-state thermal distribution around the micro-coil in brain tissue. The simulation highlights a critical maximum temperature of 41.98°C at the implant-tissue interface.

The simulation reveals a maximum localized temperature of 41.985°C at the center of the Rx coil, corresponding to a ΔT of nearly $+5^\circ\text{C}$.

From a biomedical engineering perspective, this result highlights a critical biosafety failure for continuous operation at 50 mW. The tissue adjacent to the implant would suffer severe thermal damage. To mitigate this and respect the 1°C safety threshold, future design iterations must incorporate either a highly thermally conductive biocompatible encapsulation (to distribute the heat over a larger surface area) or rely on a duty-cycled power delivery protocol rather than continuous wave transmission.

III. GENERAL CONCLUSION & PERSPECTIVES

This portfolio demonstrated a comprehensive multiphysics approach to modeling wireless neural interfaces. Through Ansys Maxwell 3D, the feasibility of magnetic power transfer to a millimeter-scale implant was successfully validated, yielding a realistic coupling coefficient ($k = 0.0116$) for a loosely coupled 15 mm transcutaneous link. However, the subsequent Ansys Icepak thermal analysis highlighted the severe biomedical limitations of continuous power delivery. The simulation revealed a localized tissue temperature spike exceeding safe limits ($\Delta T \approx +5^\circ\text{C}$ at 50 mW), which would lead to thermal necrosis.

By identifying this critical thermal bottleneck before physical prototyping, this study proves the immense value of digital twin simulations in medical device development. These findings underscore the necessity of a holistic engineering approach where electromagnetics and thermodynamics are co-optimized.